

EXPERIMENT-3

Write a program of crypt arithmetic using python code

Aim : To solve a Cryptarithmic problem using Python .

Procedure:

Assign digits to letters.

Ensure each letter has a unique digit.

Check whether the arithmetic equation is satisfied.

Display the valid solution.

Python Code:

```
from itertools import permutations

letters = ('S','E','N','D','M','O','R','Y')
digits = (0,1,2,3,4,5,6,7,8,9)

for p in permutations(digits, 8):

    S,E,N,D,M,O,R,Y = p

    if S == 0 or M == 0:

        continue

    SEND = 1000*S + 100*E + 10*N + D

    MORE = 1000*M + 100*O + 10*R + E

    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:

        print("Solution Found\n")

        print("SEND =", SEND)

        print("MORE =", MORE)

        print("MONEY =", MONEY)

        print("\nLetter Values")

        print("S =", S)
```

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print("E =", E)

print("N =", N)

print("D =", D)

print("M =", M)

print("O =", O)

print("R =", R)

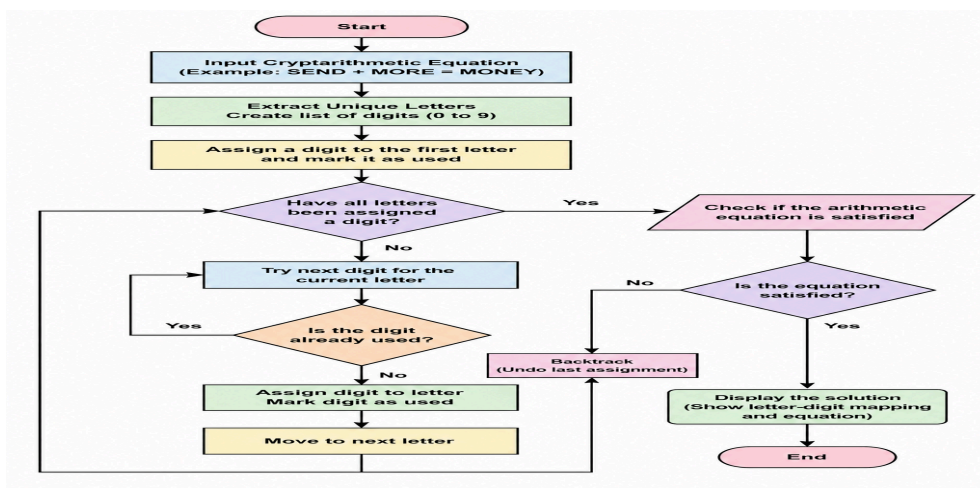
print("Y =", Y)
break

```

Output:

in.py	Output
<pre> from itertools import permutations letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y') for p in permutations(range(10), 8): S, E, N, D, M, O, R, Y = p if S == 0 or M == 0: continue SEND = 1000*S + 100*E + 10*N + D MORE = 1000*M + 100*O + 10*R + E MONEY = 10000*M + 1000*O + 100*N + 10*E + Y if SEND + MORE == MONEY: print("SEND =", SEND) print("MORE =", MORE) print("MONEY =", MONEY) break </pre>	<pre> SEND = 9567 MORE = 1085 MONEY = 10652 === Code Execution Successful === </pre>

Flowchart:



Result: Hence, crypt arithmetic problem has solved